

FEDERAL INSTITUTE OF SCIENCE AND TECHNOLOGY (FISAT)

Hormis Nagar, Mookkannoor PO, Angamaly, Kochi

Accredited by NAAC with 'A+' Grade



DEPARTMENT OF COMPUTER APPLICATIONS

SYNOPSIS OF THE MINI PROJECT

Name of the Student	Mahendran DM
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Name of the Project Guide	Dr.Deepa Mary Mathew
GitHub ID	https://github.com/MahendranDM
Project Title	RaySense
Area of the Project	AI-Based Solar Energy Monitoring, Forecasting and Performance Management
Date of Submission	20/07/2026
The growing adoption of rooftop solar installations creates a need for intelligent systems to monitor, forecast, and identify performance issues in solar energy generation. RaySense is an AI-based solar energy monitoring, forecasting, and performance management system designed for rooftop solar users. RaySense uses historical solar-generation data, panel/system specifications, and weather parameters such as temperature, humidity, cloud cover, wind speed, and solar irradiance to forecast expected solar generation using regression-based machine-learning models. During the initial development phase, a simulated solar-energy dataset will be used for model development and evaluation. Predicted and actual generation are compared to detect deviations and potential underperformance. The system provides a dashboard for solar generation and consumption monitoring, localized weather information, performance analysis, reports, alerts for significant performance deviations, and savings estimation based on applicable grid electricity rates. Consumption data is used for monitoring and cost analysis, while ML forecasting focuses on solar generation. The system consists of two main modules: a User Module providing monitoring, weather information, forecasting, performance analysis, alerts, savings estimation, and reports; and an Admin Module providing user, system, and dataset/model management. RaySense moves beyond basic solar monitoring by providing data-driven forecasting and performance insights to support better rooftop solar management.	
Front End & Back End Tools	Front End: HTML5, CSS3, JavaScript, Bootstrap, Chart.js Back End: Python, Django Database: MySQL ML: Scikit-learn, Pandas, NumPy API: OpenWeather API